We'll cover all of these terms so that you understand what each is, how they differ from each other, and how you'll interact with each over time. We're going to start at the top and work our way down, with Elixir first.

Elixir Builds on Erlang

Elixir[10] is a functional programming language that was first released in 2012. It places an emphasis on expressive syntax that conveys the meaning of code quickly.

Elixir is a compiled language. This means that the code you write will be transformed into "something else" before it runs. Ruby, on the other hand, is an interpreted language, which means that source code is directly executed by a Ruby interpreter. The Ruby interpreter might still make optimizations to your code—like apply JIT optimizations to your code—but you can save a Ruby file and run it as is.

You'll write Elixir code when you work on your app, but it's compiled into a totally different programming language called Erlang. This seems confusing at first, but it's not magic. The Elixir compiler runs a few passes over your code to turn it into Erlang code. This Erlang code gets compiled into `.beam` files that are executed by the BEAM virtual machine. You'll likely never have to worry about this unless you have very advanced deployment needs.

Due to this compilation strategy, Elixir and Erlang code can be used in the same application. You wouldn't normally do this yourself, but you'll definitely use libraries in your Elixir application that are written in Erlang. And you likely won't even know unless you look at a library's source code.

A lot of things in the Elixir ecosystem are directly built on top of what Erlang provides. Let's look at Erlang and its set of libraries called OTP.

Erlang/OTP

Erlang[11] is a functional programming language that was first released in 1998. It started its journey as a closed-source language developed in 1986 by the telecommunications company Ericsson. As such, it was designed for the challenges of telecommunications. Parallelism, fault isolation, and data isolation were built into the language runtime instead of being tacked on later.

Erlang is a programming language, but its core set of libraries is separate from it. These libraries are referred to as OTP, an acronym for Open Telecom Platform.

10. https://elixir-lang.org/
11. https://www.erlang.org/